

Final Exam - DevOps Submission (Task 2)

Student Information

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- **Group:** A

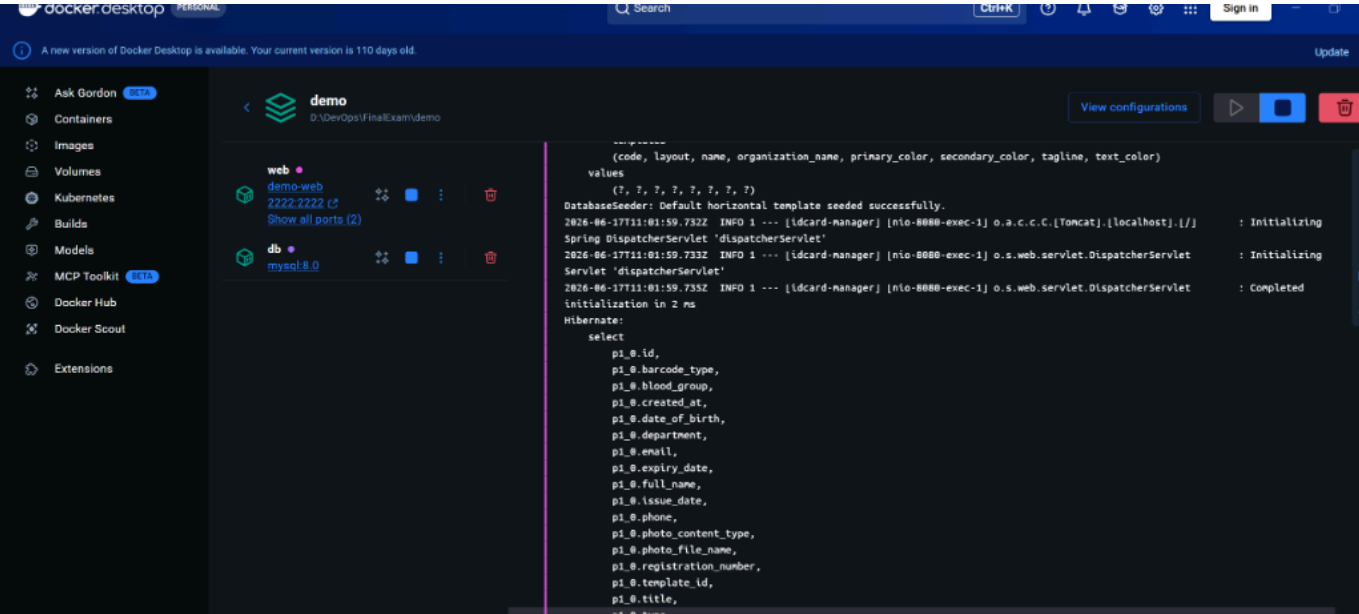
Task 2: Docker Compose Multi-Container Deployment

I have containerized the Spring Boot application and configured the MySQL database using Docker Compose (Choice A) as specified in the exam requirements.

Below are the screenshots and descriptions demonstrating the successful implementation and verification of this task.

1. Docker Desktop Overview

This screenshot shows the **demo** multi-container stack running on Docker Desktop. Both the **web** container (hosting Nginx, OpenSSH, and Spring Boot) and the **db** container (hosting MySQL) are running successfully in harmony.



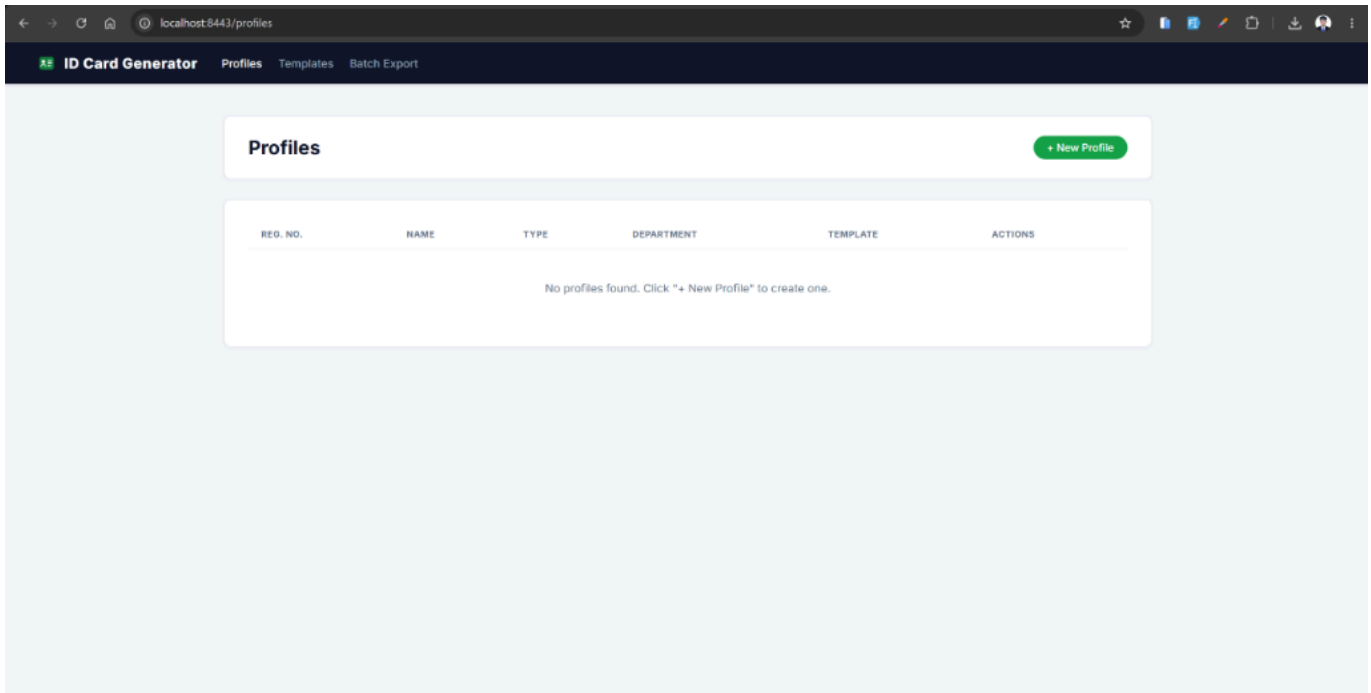
2. Docker Compose Port Expositions

The running containers list shows that port **2222** (SSH) and port **8443** (Nginx proxy) are successfully exposed and mapped to the host system.

```
PS C:\WINDOWS\system32> cd D:\DevOps\FinalExam\demo
PS D:\DevOps\FinalExam\demo> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
8d1f04d188a    mysql:8.0     "docker-entrypoint.s..." 9 minutes ago  Up 9 minutes  3306/tcp, 33060/tcp
254207c3b4bd   demo-web      "/entrypoint.sh"         9 minutes ago  Up 9 minutes  0.0.0.0:2222->2222/tcp, [::]:2222->2222/tcp, 0.0.0.0:8443->8443/tcp, [::]:8443->8443/tcp
PS D:\DevOps\FinalExam\demo>
```

3. Nginx Reverse Proxy Web Access

The web application is successfully accessed in the browser at <http://localhost:8443/profiles> via the Nginx proxy container, which forwards traffic to Spring Boot running internally on port 8080.



4. SSH Login Verification

I verified SSH connectivity to the container by running `ssh root@localhost -p 2222`. As shown below, login succeeded using the password `Hello@123`.

```
phe@DESKTOP-I228BDP:~$ ssh root@localhost -p 2222
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:QRuZRsCJBSxI0Mun6C65abbKSeUdlyI1rt0zEXJlj34.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.
root@localhost's password:
Permission denied, please try again.
root@localhost's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@254207c3b4bd:~# |
```

5. MySQL Database Verification

I verified the database name **A-PHE_Rithika-db** (Group: A, Name: PHE Rithika) and confirmed that the Spring Boot database seeder successfully initialized the **profiles** and **templates** tables inside the MySQL container.

```

1 FROM eclipse-temurin:25-jdk
2
3 # Install NGINX, OpenSSH Server, Git, and curl
4 RUN apt-get update && apt-get install -y nginx openssh-server git curl && rm -rf /var/lib/apt/lists/*
5
6 # Configure SSH
7 RUN mkdir -p /var/run/ssh
8 RUN echo 'root:Hello@123' | chpasswd
9 RUN sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config
10 # Run SSH on port 2222 inside the container
11 RUN sed -i 's/#Port 22/Port 2222/' /etc/ssh/sshd_config
12
13 # Configure NGINX to proxy port 8443 to port 8080 (Spring Boot)
14 RUN rm -f /etc/nginx/sites-enabled/default
15 COPY nginx.conf /etc/nginx/sites-enabled/default
16
17 # Set working directory
18 WORKDIR /app
19
20 # Copy the Spring Boot files from context (demo directory)
21 COPY . /app
22
23 # Build the Spring Boot application (using local Gradle wrapper inside container)
24 RUN chmod +x gradlew && ./gradlew clean bootJar -x test
25
26 # Copy entrypoint script and make it executable
27 COPY entrypoint.sh /entrypoint.sh
28 RUN chmod +x /entrypoint.sh
29
30 # Expose Nginx proxy port 8443 and SSH port 2222
31 EXPOSE 8443 2222
32
33 ENTRYPOINT ["/entrypoint.sh"]
34

```

6. Docker Compose Configuration file

This shows the **docker-compose.yml** file in the project workspace defining the service boundaries, dependencies, and environment variables.

```

1 version: "3.8"
2
3 services:
4   db:
5     image: mysql:8.0
6     container_name: db
7     restart: always
8     environment:
9       MYSQL_DATABASE: "A-PHE_Rithika-db"
10      MYSQL_ROOT_PASSWORD: "Hello@123"
11     volumes:
12       - db_data:/var/lib/mysql
13
14   web:
15     build:
16       context: .
17       dockerfile: Dockerfile
18     container_name: web
19     restart: always
20     environment:
21       SPRING_DATASOURCE_URL: "jdbc:mysql://db:3306/A-PHE_Rithika-db?createDatabaseIfNotExist=true&useSSL=false&serverTimezone=UTC&allowPublicKeyRetrieval=true"
22       SPRING_DATASOURCE_USERNAME: "root"
23       SPRING_DATASOURCE_PASSWORD: "Hello@123"
24     ports:
25       - "8443:8443"
26       - "2222:2222"
27     depends_on:
28       - db
29
30 volumes:
31   db_data:
32     driver: local

```

7. Nginx Proxy Configuration File

This shows the `nginx.conf` configuration file in the project workspace directing traffic from proxy port `8443` to the internal Spring Boot port `8080`.

```
> N nginx.conf
server {
    listen 8443 default_server;
    listen [::]:8443 default_server;

    server_name _;

    location / {
        proxy_pass http://127.0.0.1:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

Ctrl+L for Agent

Repository Details

- **Repository URL:** <https://github.com/Rithika11-ui/I4A-FirstProjectJenkins.git>
- **Branch:** `finalexam`